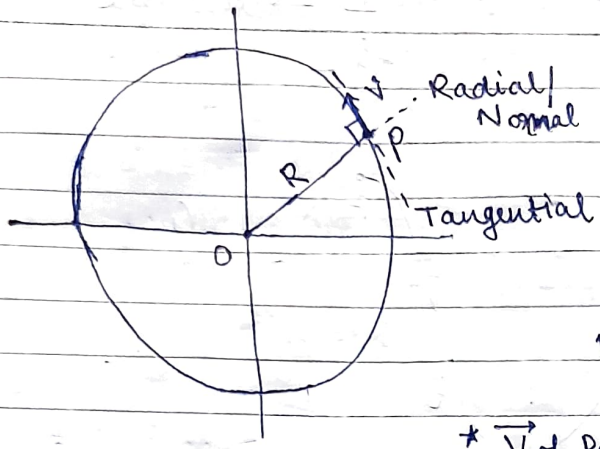
Circular Motion

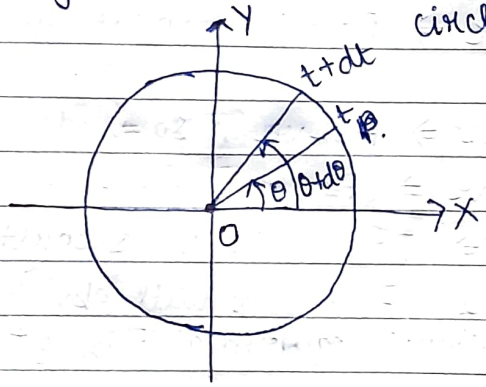


If Particle P is moving in a plane in such a way that, its distance from point 'O' remains fixed, then we say that P is performing circular motion about point 'O'.

O → Center OP → Radius

* $\vec{V}$ of Point P (wrt O), should always be ⊥ to line OP

Angular Position :- It tells the location/position of particle P on circle. Unit = Radian



θ is the angular position.

Angular displacement : Change in angular position is defined as angular displacement.

Unit = Radian

Dimensionless

→ $\vec{d\theta}$ is angular displacement

$\vec{d\theta}$ is a vector quantity

Direction generally, clockwise or anticlockwise

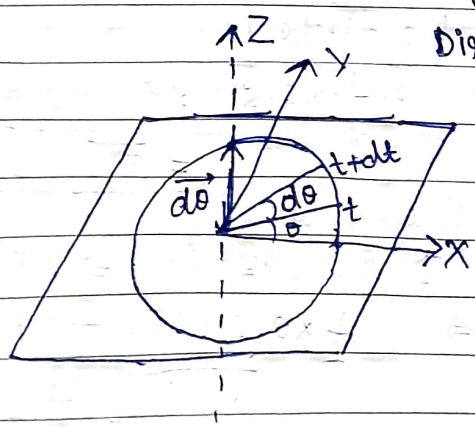
* Curl your fingers of your right hand in direction of displacement, thumb gives direction of $\vec{d\theta}$

Direction can also be represented by

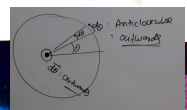
⊙ → out of plane ($\hat{k}$)

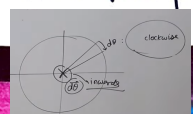
⊗ → into the plane ($-\hat{k}$)

Direction of $\vec{d\theta}$



$\vec{d\theta}$ is axial vector as ⊥ to the plane of motion.





Average angular velocity (ω):

$$\boxed{\omega_{avg} = \frac{\Delta\theta}{\Delta t} = \frac{\theta_f - \theta_i}{\Delta t}}$$

→ Unit = rad/s
→ Dimension = $[T^{-1}]$
→ Defined for finite time interval.

Time period (T) = Time taken to complete 1 Revolution.

$$\boxed{\omega_{avg} = \frac{2\pi}{T}} \quad \boxed{T = \frac{2\pi}{\omega}} \quad \text{Unit} = \text{sec}$$

Frequency (f) = No. of revolutions in 1 second

$$\boxed{f = \frac{1}{T}}$$

Unit = Sec^{-1} = Hertz (Hz)
↳ Cycle per second (C.P.S)

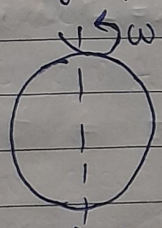
Q Find ω_{avg} of ① seconds hand ② minute hand ③ hour hand of a clock.

① Seconds Hand = $\omega_{avg} = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{60 \text{ s}} = \frac{\pi}{30}$ rad/s

② Minute hand = $\omega_{avg} = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{60\times60 \text{ s}} = \frac{\pi}{1800}$ rad/s

③ Hour hand = $\omega_{avg} = \frac{\Delta\theta}{\Delta t} = \frac{2\pi \text{ rad}}{12 \text{ hrs}} = \frac{2\pi \text{ rad}}{12\times60\times60 \text{ s}} = \frac{\pi}{21600}$ rad/s

Q Find ω_{avg} of earth's rotation about its own axis.

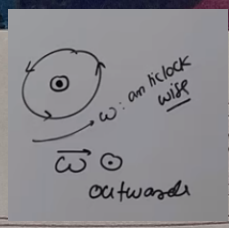


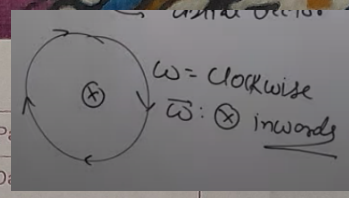
(can not be used mathematically)

$$\omega_{avg} = \frac{2\pi \text{ rad}}{24 \text{ hrs}} = \frac{\pi}{12} \text{ rad/hr}$$

Q Find ω_{avg} of earth for revolution around sun.

$$\omega_{avg} = \frac{2\pi \text{ rad}}{365.25 \text{ days}} = \frac{2\pi \text{ rad}}{365.25\times24\times60\times60 \text{ s}}$$





Instantaneous Angular Velocity :

$$\vec{\omega} = \frac{d\vec{\theta}}{dt}$$

→ Unit = rad/s , Dimension = $[T^{-1}]$
→ Vector (axial vector).
→ Direction is same as $\vec{d\theta}$.

Average Angular acceleration (α) :

$$\vec{\alpha}_{avg} = \frac{\Delta\vec{\omega}}{\Delta t} = \frac{\vec{\omega_f} - \vec{\omega_i}}{\Delta t}$$

→ Unit = Rad/sec^2
→ Dimension = $[T^{-1}]$
→ If $\vec{\omega}$ = constant $\Rightarrow \vec{\alpha} = 0 \Rightarrow$ Uniform Circular Motion (U.C.M.)

Instantaneous Angular acceleration :

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt} = \frac{d\theta}{d\theta}\cdot\frac{d\omega}{dt} = \frac{d\theta}{dt}\cdot\frac{d\omega}{d\theta}$$

$$\vec{\alpha} = \omega\frac{d\omega}{d\theta}$$

$\alpha = \frac{d\omega}{dt} = \frac{d^2\theta}{dt^2}$

→ If $\vec{\alpha}$ and $\vec{\omega}$ in same direction $\Rightarrow \omega\uparrow$
→ If $\vec{\alpha}$ and $\vec{\omega}$ in opposite direction $\Rightarrow \omega\downarrow$

Q Angular position of a particle performing circular motion is given as $\theta = (5t^3 - 10t^2 + 2t + 3)$ rad

① Find ω at any time t
② Find α at any time t.

① $\omega = \frac{d\theta}{dt} = (15t^2 - 20t + 2)$ rad/s

② $\alpha = \frac{d\omega}{dt} = (30t - 20)$ rad/s^2

Page No.
Date:

Q Angular velocity is given as $\omega = \sqrt{25+4\theta}$. Find angular acceleration.

$$\alpha = \omega \frac{d\omega}{d\theta} = \sqrt{25+4\theta} \times \frac{4}{2\sqrt{25+4\theta}} = 2 \text{ rad/sec}^2$$

Find angular acc^n
$\alpha = \omega \frac{d\omega}{d\theta}$
$\omega^2 = 25 + 4\theta$
$2\omega \frac{d\omega}{d\theta} = 0 + 4(1) = 4 = 2$
$\omega \alpha = \omega \frac{d\omega}{d\theta} = 2$ rad/sec^2

Uniform Circular Motion (UCM):

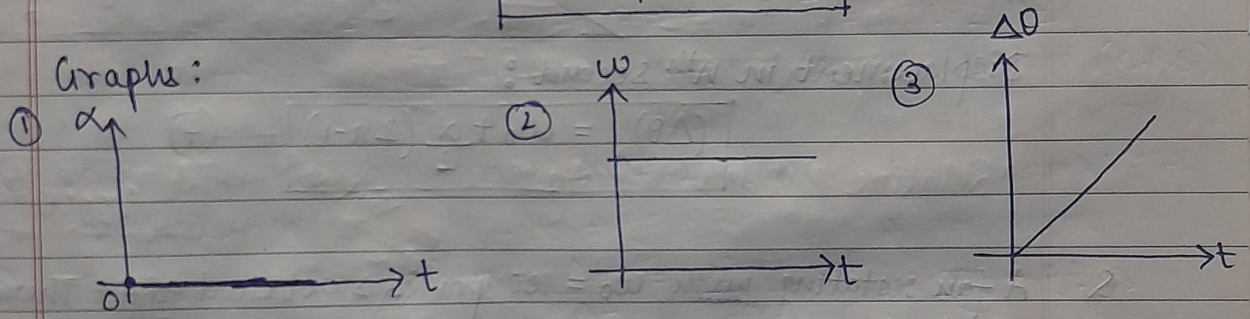
$\vec{\omega} = \text{const} \Rightarrow \vec{\alpha} = 0$

$$\frac{d\theta}{dt} = \omega$$

$$= \int_{\theta_i}^{\theta_f} d\theta = \omega \int_0^t dt$$

$$= \boxed{\Delta\theta = \theta_f - \theta_i = \omega t}$$

Graphs:



Uniform angular acceleration: $\vec{\alpha} = \text{const}$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt}$$

$$= \int_{\omega_i}^{\omega_f} d\vec{\omega} = \vec{\alpha} \int_0^t dt$$

$$= \vec{\omega_f} - \vec{\omega_i} = \vec{\alpha} t$$

$$= \boxed{\omega = \omega_0 + \alpha t} \quad \text{—①}$$

$\omega \rightarrow$ final
$\omega_0 \rightarrow$ initial

scalar form; but we must put every quantity with appropriate sign since they are vector

$$\frac{d\theta}{dt} = \omega_0 + \alpha t$$

$$= \int_{\theta_i}^{\theta_f} d\theta = \omega_0 \int_0^t dt + \alpha \int_0^t t\, dt$$

$$= \theta_f - \theta_i = \omega t + \frac{1}{2} \alpha t^2$$

$$\boxed{\Delta\theta = \omega_0 t + \frac{1}{2} \alpha t^2} \quad \text{—②}$$

Eliminate t from ① and ②:

$$\boxed{\omega^2 = \omega_0^2 + 2\alpha\theta} \quad \text{—③}$$

Eliminate ω_0 from ① and ②:

$$\boxed{\Delta\theta = \omega t - \frac{1}{2}\alpha t^2} \quad \text{—④}$$

Add eqn ② and ④:

$$\boxed{\Delta\theta = \frac{(\omega_0 + \omega)t}{2}} \quad \text{—⑤}$$

$$\boxed{\omega_{avg} = \frac{\Delta\theta}{t} = \frac{\omega_0 + \omega}{2}} \quad \text{—⑥}$$

Displacement in n^{th} second: time interval b/w t=n sec. and t=(n-1) sec.

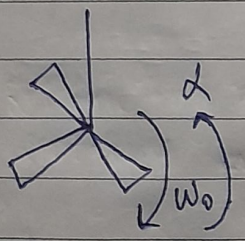
$$\boxed{(\Delta\theta)_n = \omega_0 + \frac{\alpha}{2}(2n-1)} \quad \text{—⑦}$$

Q A fan rotating with $\omega_0 = 100$ rpm is ~~switched off~~. There is constant angular retardation $\alpha = 2$ rad/sec^2. Find

① Time taken to stop:

1 rpm $=$ rotations/minute

$$= \frac{2\pi}{60}\frac{\text{rad}}{\text{s}} = \frac{\pi}{30}\text{ rad/s}$$



$$\omega_0 = 100 \times \frac{\pi}{30} = \frac{10\pi}{3}\text{ rad/s}$$

clockwise = +
anticlockwise = −

$\omega = 0$ $\quad \alpha = 2$ rad/s^2

$\omega = \omega_0 + \alpha t$

$0 = \frac{10\pi}{3} - 2t \Rightarrow \boxed{t = \frac{5\pi}{3}\text{ s}}$

② Angle travelled before it stops:

$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$

$0 = \left(\frac{10\pi}{3}\right)^2 + 2(-2)\Delta\theta \Rightarrow \boxed{\Delta\theta = \frac{100\pi^2}{36}\text{ rad}}$

③ No. of rotations :

1 rotation $\longleftrightarrow 2\pi$

1 rad $\longrightarrow \frac{1}{2\pi}$ rotations

$\Delta\theta \longrightarrow \frac{\Delta\theta}{2\pi}$ rotations.

$$\therefore \boxed{\text{No of rotations} = \frac{\Delta\theta}{2\pi} = \frac{100\pi^2}{36 \times 2\pi} = \frac{50\pi}{36}}$$

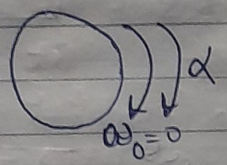
Q A disc initially at rest starts rotating with constant $\alpha = 4\ \text{rad/s}^2$

Find ① ω after 5s

② Angle travelled in these 5s

③ No. of rotations in these 5s

①



$\omega_0 = 0 \quad \alpha = 4\ \text{rad/s}^2 \quad t = 5\text{s}$

$\omega = \omega_0 + \alpha t \Rightarrow \omega = 4 \times 5$

$$\boxed{\omega = 20\ \text{rad/s}}$$

②

$\omega_0 = 0 \quad \alpha = 4\ \text{rad/s}^2 \quad t = 5\text{s}$

$\Delta\theta = \omega_0 t + \frac{1}{2}\alpha t^2 \Rightarrow \Delta\theta = \frac{1}{2} \times 4 \times 25 = 50\ \text{rad}$

③ No. of rotations $= \frac{\Delta\theta}{2\pi} = \frac{50}{2\pi} = \frac{25}{\pi}$

④ Find angle travelled in 5th second:

$(\Delta\theta)_n = \omega_0 + \frac{\alpha}{2}(2n-1)$

$(\Delta\theta)_5 = \frac{4}{2}(2(5)-1) = 18\ \text{rad}$

Q A disc has initial angular velocity $= \omega_0$ and variable angular retardation directly proportional to root of instantaneous angular velocity. Find average angular velocity of disc in the time interval in which the disc stops.

$$\omega_{avg} = \frac{\text{Angle travelled before disc stops}}{\text{Time taken by disc to stop}} = \frac{\Delta\theta}{t}$$

$\alpha \propto \sqrt{\omega}$

$\Rightarrow \alpha = K\sqrt{\omega} = -\frac{d\omega}{dt}$

$= K\sqrt{\omega} = -\frac{d\omega}{dt}$

$= -K\,dt = \frac{d\omega}{\sqrt{\omega}}$

$= -K\int_0^t dt = \int_{\omega_0}^{0} \frac{d\omega}{\sqrt{\omega}}$

$= +Kt = +2\sqrt{\omega_0}$

$= \boxed{t = \frac{2\sqrt{\omega_0}}{K}}$

$\alpha = K\sqrt{\omega} = -\omega\frac{d\omega}{d\theta}$

$= K\sqrt{\omega} = -\omega\frac{d\omega}{d\theta}$

$= -K\int_0^{\Delta\theta} d\theta = \int_{\omega_0}^{0} \sqrt{\omega}\, d\omega$

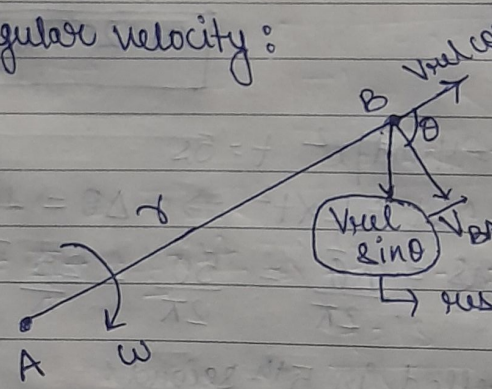
$= +K\Delta\theta = +\frac{2}{3}\omega_0^{3/2}$

$\boxed{\Delta\theta = \frac{2\omega_0^{3/2}}{3K}}$

$\omega_{avg} = \frac{\Delta\theta}{t} = \frac{\frac{2}{3}\frac{\omega_0^{3/2}}{K}}{\frac{2\sqrt{\omega_0}}{K}} = \frac{\omega_0}{3}$ $\quad \therefore \boxed{\omega_{avg} = \frac{\omega_0}{3}}$

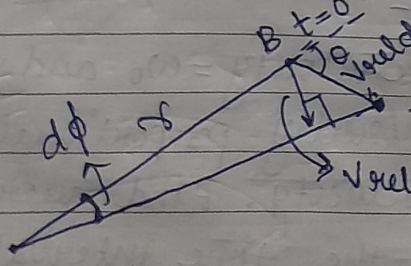
Relative angular velocity:

$\boxed{V_{rel}\cos\theta = V_{sep}}$



↳ responsible for angular velocity

Relative angular velocity (ω) is rate at which line AB is rotating in space.

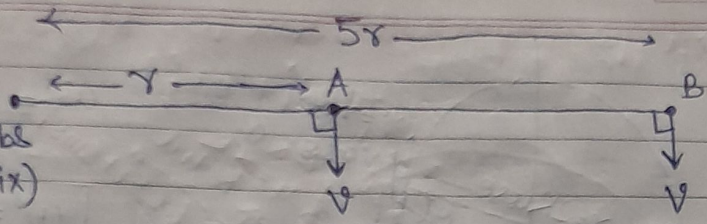


$d\phi = \frac{\text{arc length}}{\text{radius}}$

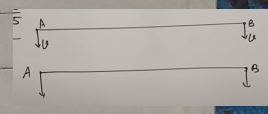
$d\phi = \frac{V_{rel}dt\sin\theta}{\gamma}$

$\boxed{\omega = \frac{d\phi}{dt} = \frac{V_{rel}\sin\theta}{\gamma}}$ $\quad \boxed{\omega = \frac{(V_{rel})_\perp}{\gamma}}$

Page No.
Date:



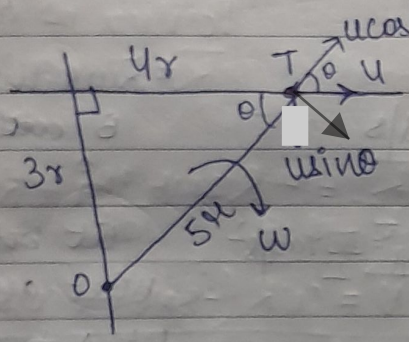
$\omega_{AO} = \frac{v}{r}$ $\quad \omega_{BO} = \frac{v}{5r}$ $\quad \omega_{BA} = \frac{(v_{BA})_\perp}{4r} = 0$



$\boxed{\omega_{BO} = \frac{\omega_{AO}}{5}}$ $\quad \therefore$ Relative angular velocity varies inversely with distance as well.

$\Rightarrow \omega \uparrow r \downarrow$, $\omega \downarrow r \uparrow$

Q

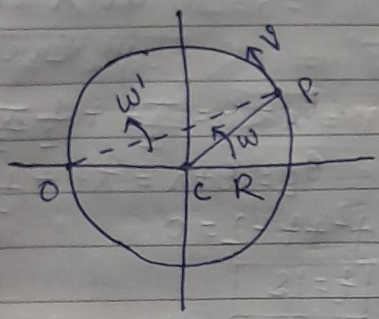


Find ω_{TO}.

$\omega = \frac{(v_{rel})_\perp}{r} = \frac{u\sin\theta}{5r}$

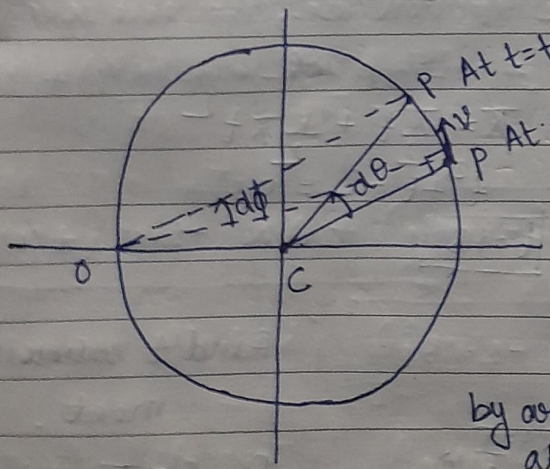
$\omega = \frac{v}{5r}\left(\frac{3r}{5r}\right) = \frac{3v}{25r}$

**Q



Angular velocity about center C is ω. Find ω' about a point on circumference.

$\omega = \frac{(v_{rel})_\perp}{r} = \frac{v}{R}$ $\quad \boxed{\omega = \frac{v}{R}}$



$\omega = \frac{d\theta}{dt}$

$\omega' = \frac{d\phi}{dt}$

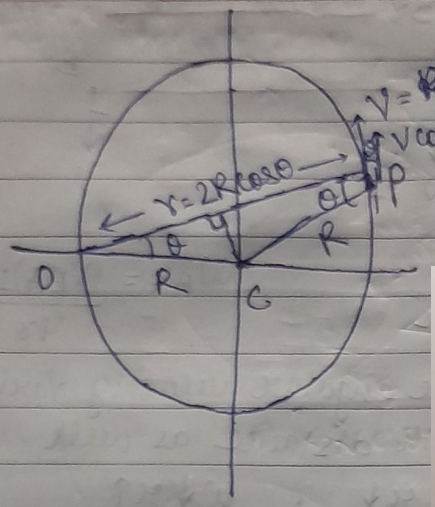
We know, $d\phi = \frac{d\theta}{2}$ as angle subtended by arc at centre is twice of angle subtended at circumference.

$\omega' = \frac{d\phi}{dt} = \frac{d\theta}{2\,dt} = \frac{\omega}{2}$ $\quad \therefore \boxed{\omega' = \frac{\omega}{2}}$

Concept

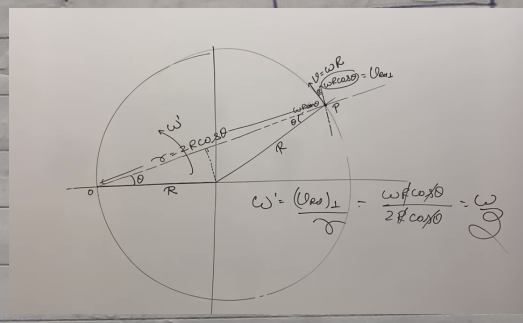
★ If a particle as angular vel and acc^n about center as ω and α then angular vel and acc^n about circumference is $\omega/2$ and $\alpha/2$

Alternate

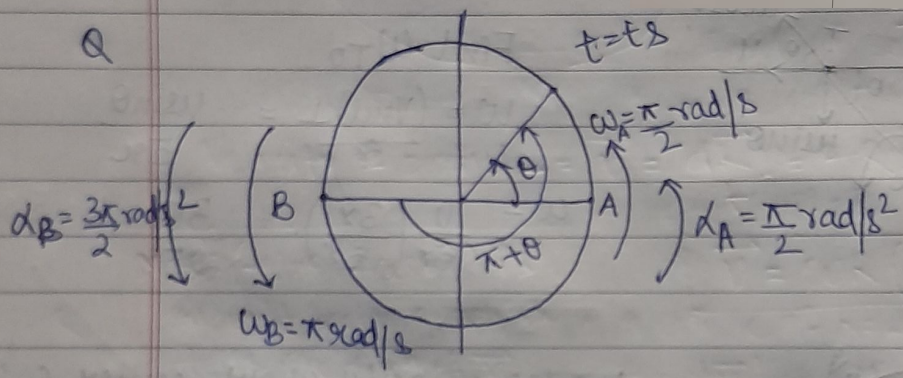


$$\omega' = \frac{R\omega\cos\theta}{2R\cos\theta}$$

$$\boxed{\omega' = \frac{\omega}{2}}$$



Q



Find the time when B catches A.

Let both of them meet after time t.
Let A travel θ displacement in time t.

$$\theta_A = \theta = \frac{\pi}{2}t + \frac{1}{2}\frac{\pi}{2}t^2 \quad -(1)$$

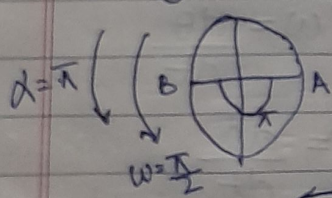
$$\theta_B = \theta + \pi = \pi t + \frac{1}{2}\frac{3\pi}{2}t^2 \quad -(2)$$

$$(2) - (1) \Rightarrow \pi = \frac{\pi}{2}t + \frac{1}{2}\pi t^2 \Rightarrow t^2 + t - 2 = 0$$

$$\boxed{t = 1s}$$

Alternate

Bring A at rest

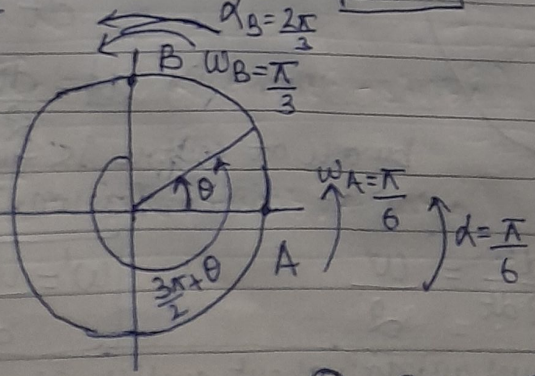


$$\pi = \frac{\pi}{2}t + \frac{1}{2}\pi t^2$$

$$t^2 + t - 2 = 0$$

$$\boxed{t = 1s}$$

Q



Find t when A and B meet.

$$\theta_A = \theta = \frac{\pi}{6}t + \frac{1}{2}\frac{\pi}{6}t^2$$

$$\theta_B = \frac{3\pi}{2} + \theta = \frac{\pi}{3}t + \frac{1}{2}\frac{2\pi}{3}t^2$$

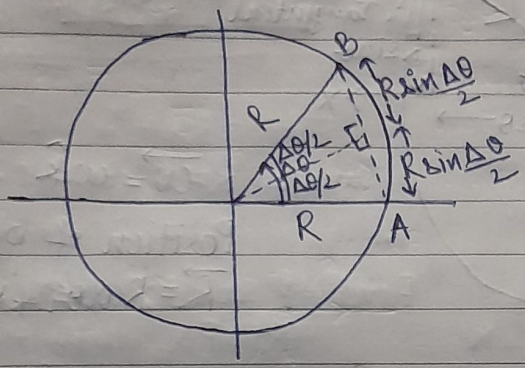
$$(2) - (1) \Rightarrow \frac{3\pi}{2} = \frac{\pi}{6}t + \frac{1}{2}\frac{\pi}{2}t^2$$

alternatively, can also be solved using relative like in previous case

$\Rightarrow \frac{3}{2} = \frac{t}{6} + \frac{t^2}{4} \Rightarrow 3t^2 + 2t - 18 = 0$

$$t = \frac{-2 + \sqrt{4 + 4 \times 3 \times 18}}{2 \times 3}$$

Relation between linear and angular terms:

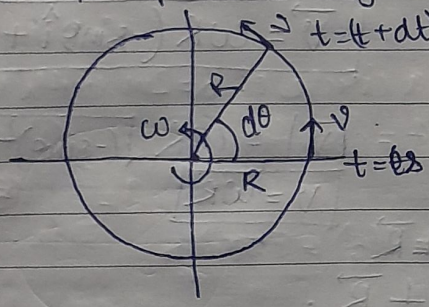


$\Delta\theta$ = Angular displacement

Distance travelled = $\overset{\frown}{AB}$
$= (\Delta\theta)R$

Displacement = AB
$= 2R\sin\left(\frac{\Delta\theta}{2}\right)$

i) Relation b/w speed and angular speed:



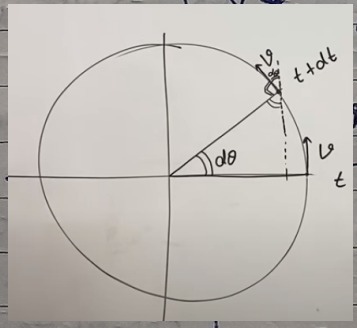
$d\theta = \frac{V\,dt}{R}$ $\left\{\theta = \frac{\text{arc}}{\text{radius}}\right\}$

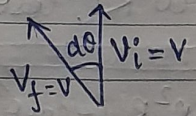
$\frac{d\theta}{dt} = \frac{V}{R}$

$\omega = \frac{V}{R}$ $\quad V = R\omega$

Acceleration in U.C.M.:

In U.C.M. $\vec{\omega} = \text{const}$, $V = \text{speed} = R\omega = \text{const}$ but $\vec{v}$ (velocity) is not constant and direction of velocity changes. Hence we get an acceleration. This accn is not responsible for change in speed, it is only responsible for change in dirn of velocity.





Change in velocity

$= dv = |\vec{V_f} - \vec{V_i}|$

$= \sqrt{v^2 + v^2 - 2v^2\cos d\theta}$

$= \sqrt{2v^2(1 - \cos d\theta)}$

$= \sqrt{2v^2\left(2\sin^2\frac{d\theta}{2}\right)}$

$= 2v\sin\frac{d\theta}{2}$ $\quad (\sin x \approx x$ x is small)

$= 2v\frac{d\theta}{2} = v\,d\theta$

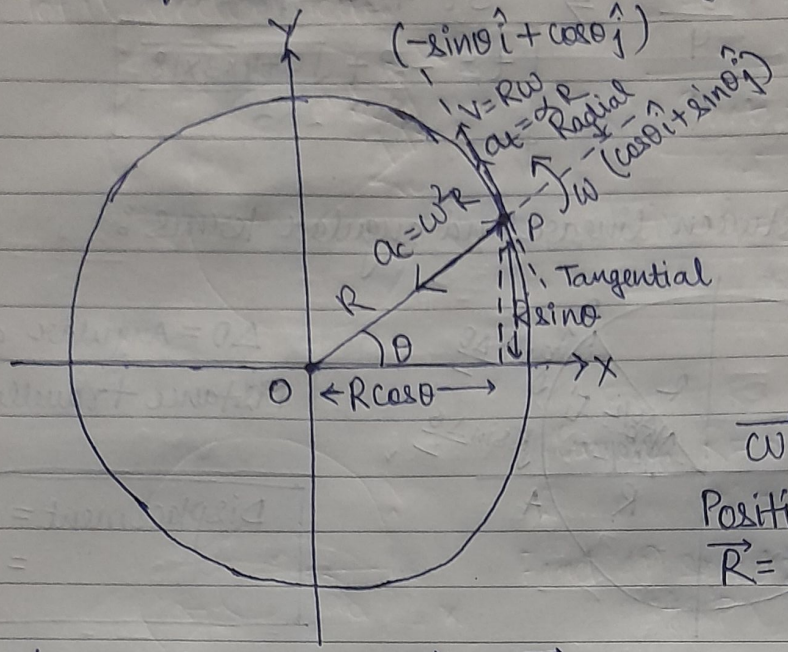
$$a_c = \frac{dv}{dt} = \frac{v\,d\theta}{dt} = v\omega$$

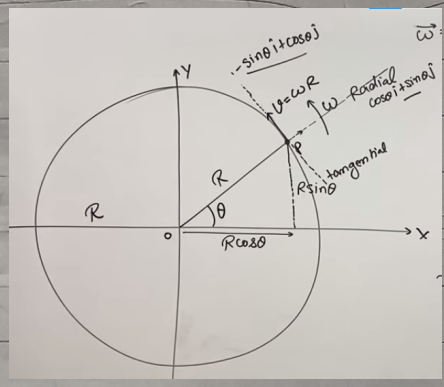
$$a_c = v\omega = \frac{v^2}{R} = \omega^2 R$$

★ This accn is known as centripetal or radial accn and acts always towards center

Page No.
Date:

Relation b/w angular and linear terms (Vectors method):





$\vec{\omega} = \omega \hat{k}$

Position of particle:

$\vec{R} = R\cos\theta\,\hat{i} + R\sin\theta\,\hat{j} = R(\cos\theta\,\hat{i} + \sin\theta\,\hat{j})$

Velocity of particle $= \vec{v} = \frac{d\vec{R}}{dt} = -R\sin\theta\frac{d\theta}{dt}\hat{i} + R\cos\theta\frac{d\theta}{dt}\hat{j}$

Note: $(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})$ is an unit vector which tells the dirⁿ of $\vec{v}$

$\boxed{\vec{v} = \omega R(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})}$ $\left[\frac{d\theta}{dt} = \omega\right]$

$|\vec{v}| = \omega R\sqrt{(-\sin\theta)^2 + (\cos\theta)^2}$

$\boxed{v = \omega R}$

* Also, $\vec{v}\cdot\vec{R} = 0 \Rightarrow \vec{v} \perp \vec{R}$.
Hence velocity is always $\perp \vec{R}$.

Acceleration of particle (General case, ω may not be constant):

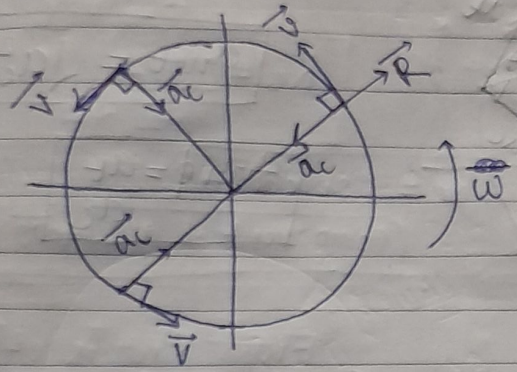
$$\vec{a} = \frac{d\vec{v}}{dt} = R\left\{\omega\left(-\cos\theta\frac{d\theta}{dt}\hat{i} - \sin\theta\frac{d\theta}{dt}\hat{j}\right) + (-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})\frac{d\omega}{dt}\right\}$$

$$\boxed{\vec{a} = \omega^2 R(-\cos\theta\,\hat{i} - \sin\theta\,\hat{j}) + \alpha R(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})}$$

$\boxed{\vec{a}_c = \omega^2 R(-\cos\theta\,\hat{i} - \sin\theta\,\hat{j})}$ (Radially inwards)

$\boxed{\vec{a}_t = \alpha R(-\sin\theta\,\hat{i} + \cos\theta\,\hat{j})}$ (Tangential, towards $\vec{v}$ if α = +ve, or opposite to $\vec{v}$ if α = -ve)

Centripetal / Radial / Normal acceleration :



$$\vec{v} = \vec{\omega} \times \vec{R}$$

$$a_c = \frac{v^2}{R} = \omega^2 R = \omega v$$

$$\vec{a_c} = \vec{\omega} \times \vec{v}$$

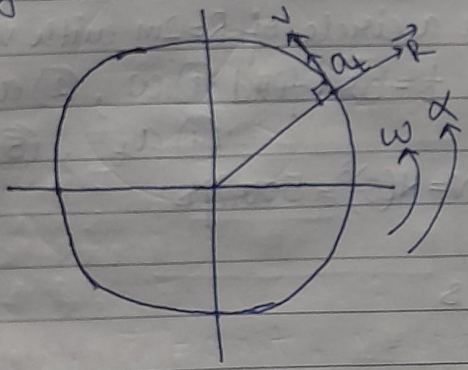
$$\vec{a_c} = \vec{\omega} \times (\vec{\omega} \times \vec{R})$$

Vector triple product.

- → It is always directed towards center.
- → $\vec{a_c} \perp \vec{v}$
- → It cannot change speed
- → It changes direction of velocity
- → In U.C.M., $|\vec{a_c}| = \text{const}$ but $\vec{a_c} \neq \text{const}$

the centripetal acceleration vector can never be constant

Tangential acceleration :



$$a_t = \alpha R$$

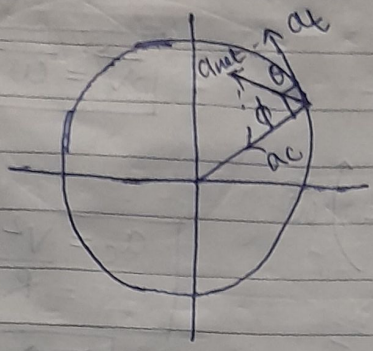
$$\vec{a_t} = \vec{\alpha} \times \vec{R}$$

- → Always along Tangential direction
- → If $\vec{a_t}$ in same direction of $\vec{v} \Rightarrow v\uparrow$
- → If $\vec{a_t}$ in opposite direction to $\vec{v} \Rightarrow v\downarrow$
- → $a_t = \frac{d|\vec{v}|}{dt}$ $\qquad a_t = \frac{d(\text{speed})}{dt}$

∴ Tangential acceleration is rate of change of speed.

- → In U.C.M, $\omega = \text{const} \Rightarrow \alpha = 0$
  $v = \text{const} \Rightarrow a_t = 0$

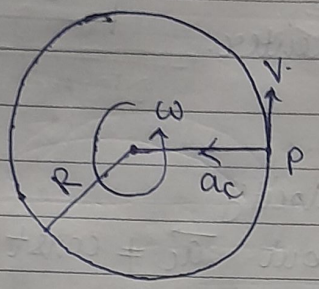
Total acceleration:



$$a_{net} = \sqrt{a_c^2 + a_t^2}$$

$$\tan\phi = \frac{a_t}{a_c} \qquad \tan\theta = \frac{a_c}{a_t}$$

Q



Given, $R = 20\,cm = 0.2\,m$

$\omega = 20\ rad/s = const$

Find ① speed of P; $v = R\omega = 0.2 \times 20 = 4\ m/s$

② acceleration of pt. P: $a_c = \omega^2 R = 400 \times 0.2 = 80\ m/s^2$

Q A particle is moving on a circle of $R = 2m$ with variable speed $v = 4t^3\ m/s$. At time $t = 5s$, find ① ω, ② a_c, ③ α, ④ a_t, ⑤ a_{net}.

① $\omega = \frac{v}{R}$, At $t = 5s$, $v = 4(5)^3 = 500\ m/s$

$\omega = \frac{500}{2} = 250\ rad/s$

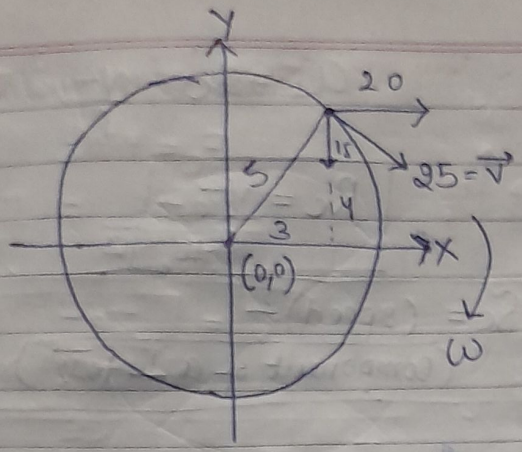
② $a_c = \frac{v^2}{R} = \frac{500 \times 500}{2} = 250 \times 500 = 125000\ m/s^2$

③ $\alpha = \frac{d\omega}{dt} = \frac{d}{dt}\left(\frac{4t^3}{2}\right) = \frac{d}{dt}(2t^3) = 6t^2 = 6 \times 5^2 = 150\ rad/s^2$

④ $a_t = \alpha R = 150 \times 2 = 300\ m/s^2$

⑤ $a_{net} = \sqrt{a_c^2 + a_t^2} = \sqrt{(125000)^2 + (300)^2}\ m/s^2$

Q



At an instant position of particle performing UCM is $(3\hat{i}+4\hat{j})$ m and its velocity at this instant is $(20\hat{i}+V_y\hat{j})$ m/s.

① Find V_y.

$\vec{V} \perp \vec{R} \Rightarrow \vec{V}\cdot\vec{R}=0$

$(20\hat{i}+V_y\hat{j})\cdot(3\hat{i}+4\hat{j})=0 \Rightarrow 60+4V_y=0$

$\therefore \boxed{V_y=-15}$ $\quad \therefore \vec{V}=20\hat{i}-15\hat{j}$

② Find ω.

$$\omega=\frac{V}{R}=\frac{\sqrt{(20)^2+(-15)^2}}{5}=\frac{25}{5}=5 \text{ rad/s}$$

$\vec{\omega}=5(-\hat{k})$ rad/s

③ Find a_c on particle.

$a_c=\omega^2 R=25\times5=125$ m/s^2

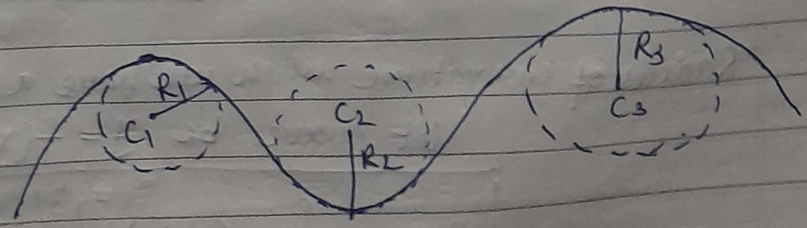
④ Find $\vec{a_c}$ at given instant.

$\vec{a_c}=\vec{\omega}\times\vec{v}=(-5\hat{k})\times(20\hat{i}-15\hat{j})=(-100\hat{j}-75\hat{i})$ m/s^2

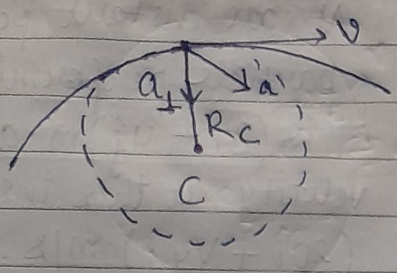
Radius of Curvature : If a particle is moving along a curved path, then the curve can be assumed to be part of imaginary circle.

→ Center of circle is center of curvature

→ Radius of circle is known as radius of curvature



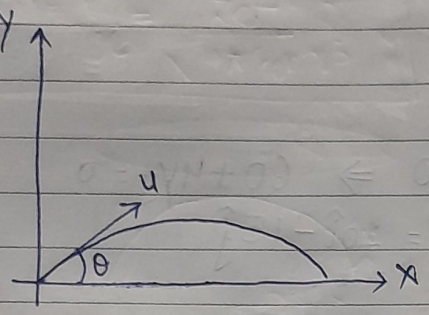
Page No.
Date:



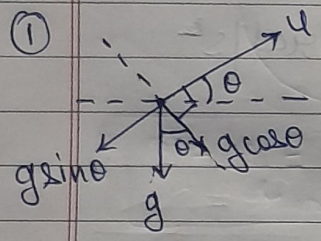
$$a_\perp = a_c = \frac{v^2}{R_c}$$

$$R_c = \frac{v^2}{a_\perp}$$

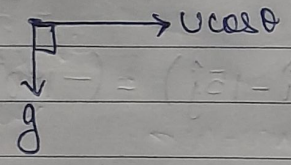
$$R_c = \frac{(\text{speed})^2}{(\text{component of } a \perp \text{ to } \vec{v})}$$

Q 

Find Radius of curvature.
① Just at the time of projection
② when ball is at max^m height
③ when ball is moving at an angle α with the horizontal

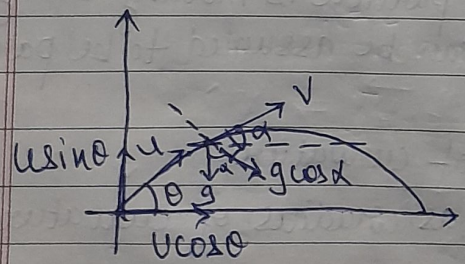
① 

$$R_c = \frac{u^2}{g\cos\theta}$$

② 

$$R_c = \frac{(u\cos\theta)^2}{g}$$

This must be $R_{c\,min^m}$ as $u\cos\theta$ is min^m velocity particle can have and g is max^m acc^n.

③ 

$$v\cos\alpha = u\cos\theta \Rightarrow v = \frac{u\cos\theta}{\cos\alpha}$$

$$R_c = \frac{u^2\cos^2\theta}{g\cos^3\alpha}$$

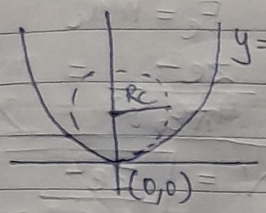
Radius of curvature when we have a curve:

Let curve be $y = f(x)$

$$R_c = \frac{\left\{1+\left(\frac{dy}{dx}\right)^2\right\}^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}$$

Page No.
Date:

Q Given parabola $y=ax^2$. Find R_c at $(0,0)$.



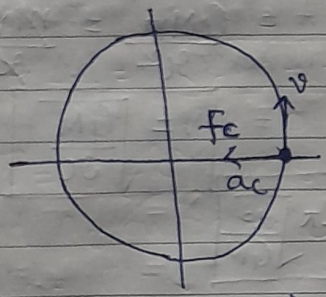
$$y=ax^2 \Rightarrow \frac{dy}{dx}=2ax \Rightarrow \frac{d^2y}{dx^2}=2a$$

$$R_c=\frac{\left\{1+\left(\frac{dy}{dx}\right)^2\right\}^{3/2}}{\left|\frac{d^2y}{dx^2}\right|}=\frac{(1+4a^2x^2)^{3/2}}{2a}$$

$$\therefore R_c \text{ at } (0,0)=\frac{(1+4a^2(0))^{3/2}}{2a}=\frac{1}{2a}$$

Dynamics of Circular Motion: **in circular motion, it is always beneficial to write forces and acceleration along the radial and tangential direction**

Centripetal Force:



$$\boxed{F_c=ma_c} \qquad \boxed{F_c=\frac{mv^2}{R}=m\omega^2R=m\omega v}$$

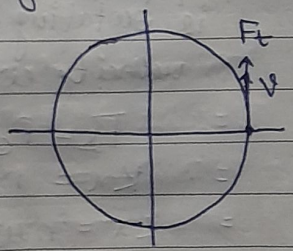
$$\boxed{\vec{F_c}=m(\vec{\omega}\times\vec{v})}$$

Power of $\vec{F_c}=\vec{F_c}\cdot\vec{v}=F_cv\cos 90^\circ=0$

Power of $\vec{F_c}=0 \Rightarrow$ W.d by $\vec{F_c}=0$

$\Rightarrow \vec{F_c}$ doesn't change speed or Kinetic energy.

Tangential Force:



$$\boxed{F_t=ma_t} \qquad \boxed{F_t=m\alpha R}$$

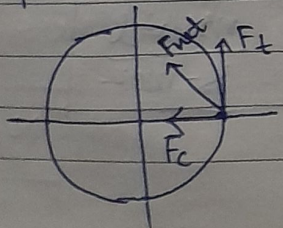
$$\boxed{F_t=m\frac{d|\vec{v}|}{dt}} \qquad \boxed{\vec{F_t}=m(\vec{\alpha}\times\vec{R})}$$

→ F_t is always tangential to the path

If $\vec{F_t}\parallel\vec{v} \Rightarrow$ speed ↑, If $\vec{F_t}\nparallel\vec{v} \Rightarrow$ speed ↓

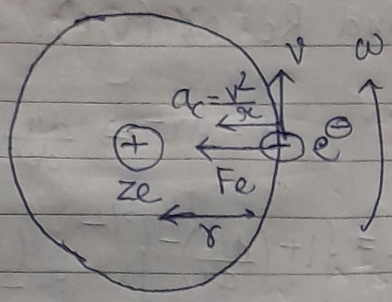
→ In UCM, speed = const $\Rightarrow \vec{F_t}=0$.

Net force:



$$\boxed{F_{net}=\sqrt{F_c^2+F_t^2}}$$

Q



Find v, ω and Time period

$F_e = ma_c$

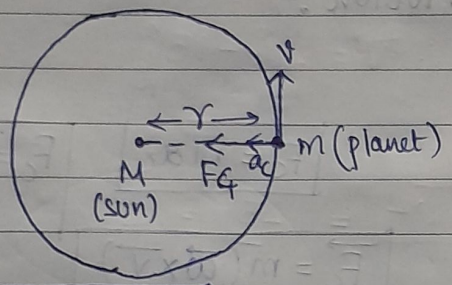
$\frac{ze^2}{4\pi\varepsilon_0 r^2} = \frac{mv^2}{r}$

$$v = \sqrt{\frac{ze^2}{4\pi\varepsilon_0 mr}}$$

$$\omega = \frac{v}{r} = \sqrt{\frac{ze^2}{4\pi\varepsilon_0 mr^3}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{4\pi\varepsilon_0 mr^3}{ze^2}}$$

Q



Find v, ω and T

$F_G = ma_c$

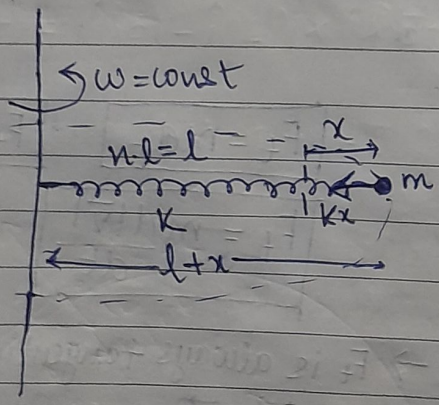
$= \frac{GMm}{r^2} = \frac{mv^2}{r}$

$= v = \sqrt{\frac{GM}{r}}$ (orbital velocity)

$$\omega = \frac{v}{r} = \sqrt{\frac{GM}{r^3}}$$

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r^3}{GM}}$$

Q Gravity free space



Find extension in spring if particle of mass m is rotating with angular velocity ω.

$F_s = F_c$

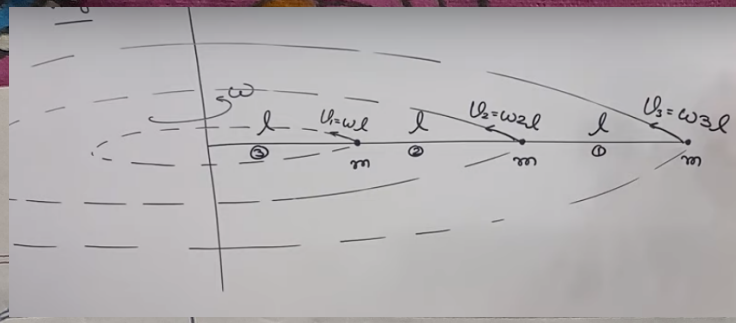
$= Kx = m\omega^2(l+x)$

$= Kx = m\omega^2 l + m\omega^2 x$

$= Kx - m\omega^2 x = m\omega^2 l$

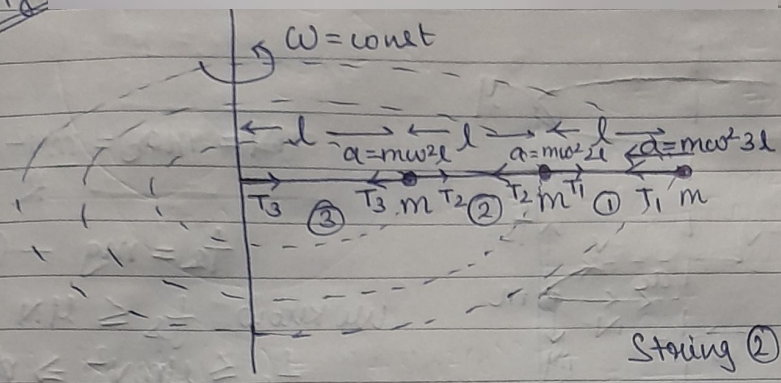
$= x(K - m\omega^2) = m\omega^2 l$

$$x = \frac{m\omega^2 l}{K - m\omega^2}$$



Page No.
Date:

Q No gravity



Find tensions in strings

String ①

$T_1 = mac$

$\boxed{T_1 = m\omega^2(3l)}$

String ②

$T_2 - T_1 = mac$

$T_2 - T_1 = m\omega^2(2l)$

$T_2 - 3m\omega^2 l = 2m\omega^2 l$

$\boxed{T_2 = 5m\omega^2 l}$

String ③

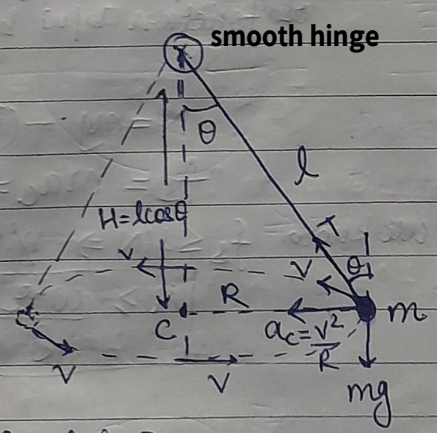
$T_3 - T_2 = mac$

$T_3 - T_2 = m\omega^2 l$

$T_3 - 5m\omega^2 l = m\omega^2 l$

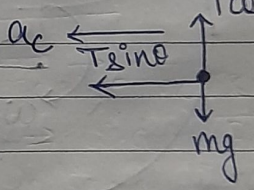
$\boxed{T_3 = 6m\omega^2 l}$

Conical Pendulum :



$R = l\sin\theta$

$\vec{V} = V(-\hat{k})$, $F_T = 0 \Rightarrow a_t = 0 \Rightarrow$ V = const



$T\cos\theta = mg$ — ①

$T\sin\theta = \frac{mv^2}{R}$ — ②

Divide ① and ②

$\tan\theta = \frac{v^2}{gR}$

$\boxed{\omega = \frac{V}{R} = \frac{V}{l\sin\theta} = \sqrt{\frac{g}{l\cos\theta}} = \sqrt{\frac{g}{H}}}$

$\boxed{V = \sqrt{gR\tan\theta}}$

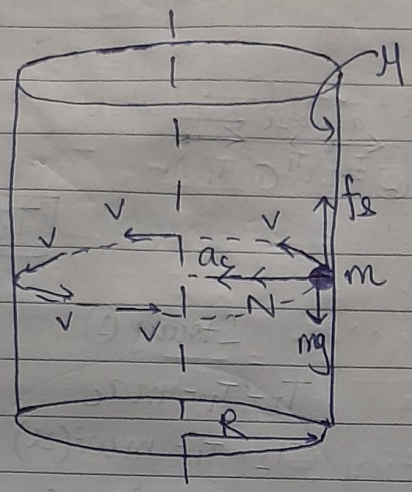
$\boxed{V = \sqrt{\frac{lg\sin^2\theta}{\cos\theta}}}$

$\boxed{\text{Time period } (t) = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{H}{g}}}$

Note: In circular motion, always try to take components in tangential and radial dirn

Page No.
Date:

Well of Death:



$$N = ma_c = \frac{mv^2}{R} \quad -(1)$$

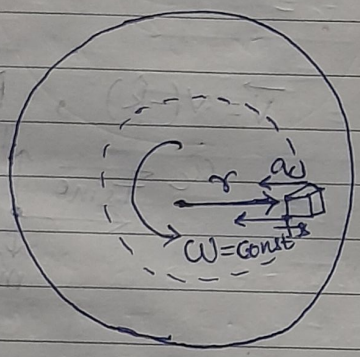
$$f_s = mg \quad -(2)$$

We know $f_s \le \mu N$

$$f_s \le \frac{\mu m v^2}{R} \Rightarrow mg \le \frac{\mu m v^2}{R}$$

$$\boxed{v \ge \sqrt{\frac{Rg}{\mu}}}$$

Q Top view



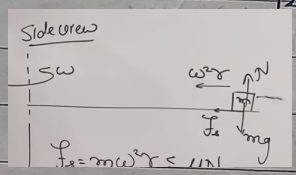
A block of mass m is rotating without slipping. Find coefficient of friction b/w block and table.

$$N = mg \quad -(1)$$

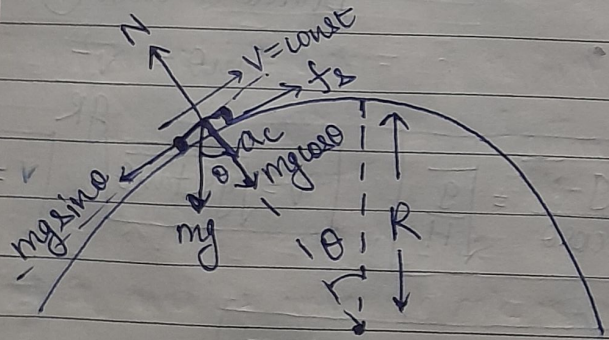
$$f_s = ma_c = m\omega^2 r \quad -(2)$$

We know $f_s \le \mu N \Rightarrow m\omega^2 r \le \mu mg$

$$\boxed{\mu \ge \frac{\omega^2 r}{g}}$$



Q Ex



Tangential direction:

$V = \text{const} \Rightarrow a_t = 0 \Rightarrow F_t = 0$

$$\therefore \boxed{f_s = mg\sin\theta}$$

Radial direction:

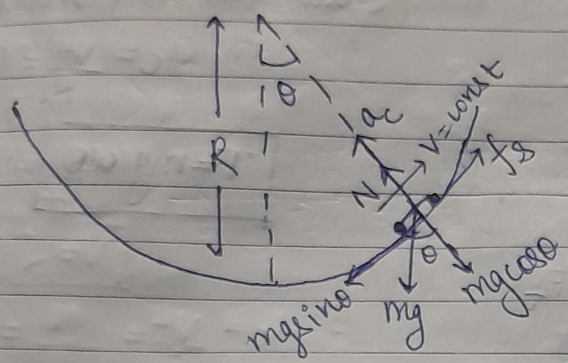
$$mg\cos\theta - N = \frac{mv^2}{R}$$

$$\boxed{N = mg\cos\theta - \frac{mv^2}{R}}$$

$$\therefore \boxed{N \propto \frac{1}{v}} \quad N \uparrow v \downarrow$$

Condition to fly off $\Rightarrow N = 0$

* Ex



Tangential: $V = \text{const} \Rightarrow F_t = 0$

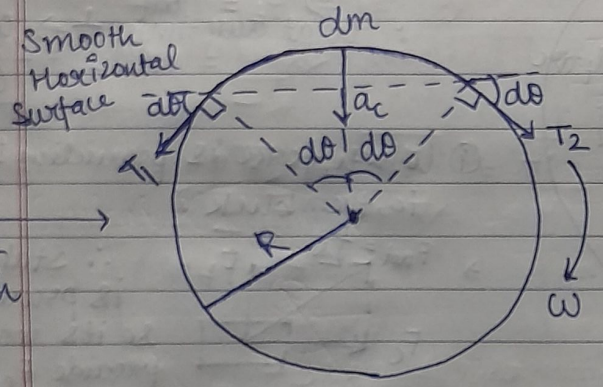
$\therefore \boxed{f_s = mg\sin\theta}$

Radial: $N - mg\cos\theta = \frac{mv^2}{R}$

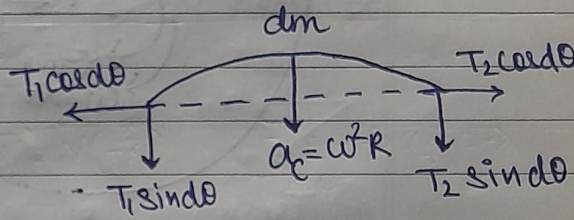
$\boxed{N = \frac{mv^2}{R} + mg\cos\theta}$

* Bike can't fly off in this case as N can't be 0.

** Q Smooth Horizontal surface



Find tension in chain.



$2\pi \longrightarrow m$

$1\,\text{rad} \longrightarrow \frac{m}{2\pi}$

$2d\theta = \frac{m}{2\pi}(2d\theta) = \frac{m}{\pi}d\theta = dm$

$T_1\cos d\theta = T_2\cos d\theta$

$\therefore T_1 = T_2 = T$ — (1)

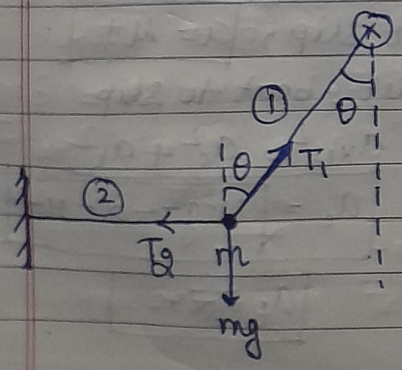
$T_1\sin d\theta + T_2\sin d\theta = dm\,\omega^2 R$

$2T\sin d\theta = \frac{m}{\pi}d\theta\,\omega^2 R$

$2T\,d\theta = \frac{m}{\pi}d\theta\,\omega^2 R$

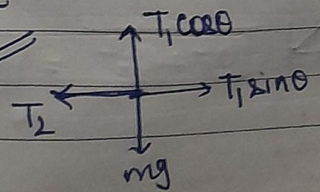
$\boxed{T = \frac{m\omega^2 R}{2\pi}}$

Q



String ② is cut.
Find tension in string ① before and just after cut.

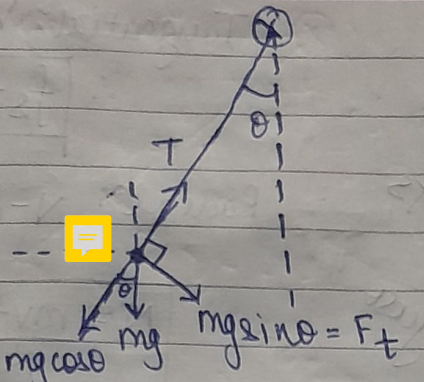
Before cut



$T_1\cos\theta = mg$

$\boxed{T_1 = mg\sec\theta}$

Just After cut

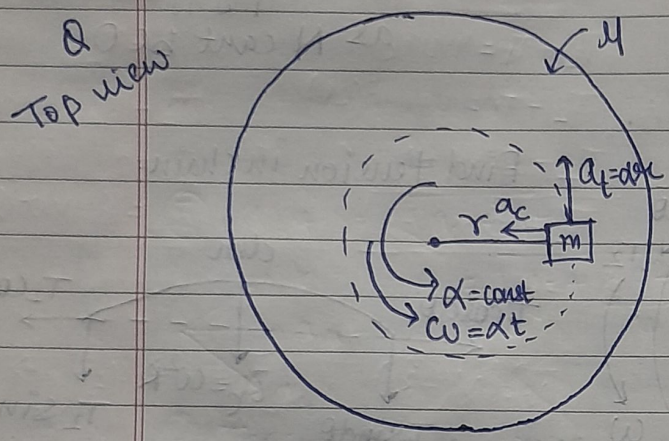


Just after cut, velocity of particle = 0

$\therefore a_c = \frac{v^2}{R} = 0$

$\boxed{T = mg\cos\theta}$

Q



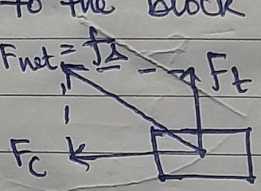
Given: $\omega_0 = 0$

α = const

$\therefore$ ω at any time $t \Rightarrow \omega = \omega_0 + \alpha t$

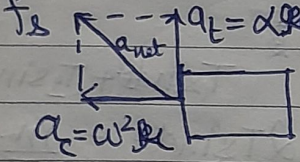
$\boxed{\omega = \alpha t}$

① Which force is providing F_c and F_t to the block?



$\Rightarrow$ $\therefore$ Static friction is providing F_{net} so its component will provide F_c and F_t.

② Find value of static friction acting on block, when block is about to slip on table.



As $t \uparrow \Rightarrow \omega \uparrow \Rightarrow a_c \uparrow \Rightarrow F_c \uparrow \Rightarrow F_{net} \uparrow \Rightarrow f_s \uparrow$

$f_{s\,max} = f_{lim} = \mu mg$

$\therefore$ when block is abt to slip, $\boxed{f_s = \mu mg}$

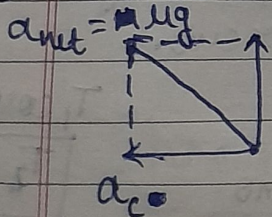
③ Find acc^n of block when it is just abt to slip on table.

$f_{s\,max} = m a_{net}$

$\mu mg = m a$

$\therefore$ when block is just abt to slip, $\boxed{a = \mu g}$

④ Find a_c and a_t when block is just about to slip.



$a_{net}^2 = a_c^2 + a_t^2$

$\boxed{a_c = \sqrt{\mu^2 g^2 - \alpha^2 r^2}}$

$\boxed{a_t = \alpha r}$

⑤ Find angular velocity of table when block is just about to slip on table.

$$a_c = \omega^2 r = \sqrt{\mu^2 g^2 - \alpha^2 r^2}$$

$$\boxed{\omega = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{r^2}\right)^{1/4}}$$

⑥ Find the time when block is just about to slip.

$$\omega = \omega_0 + \alpha t$$

$$\omega = \alpha t \Rightarrow t = \frac{\omega}{\alpha} = \frac{\left(\frac{\mu^2 g^2 - \alpha^2 r^2}{r^2}\right)^{1/4}}{\alpha}$$

$$\boxed{t = \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{\alpha^4 r^2}\right)^{1/4}}$$

⑦ Find the angle travelled by block before slipping.

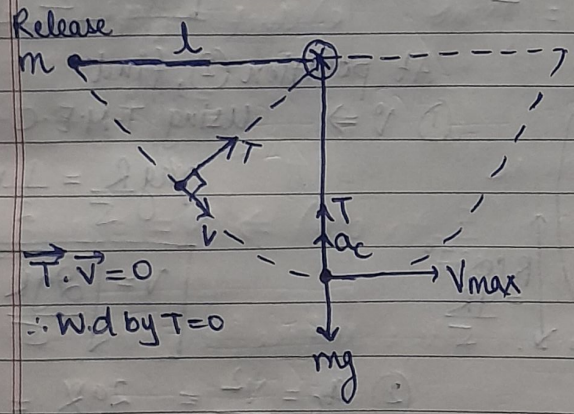
$$\omega^2 = \omega_0^2 + 2\alpha\Delta\theta$$

$$= \left(\frac{\mu^2 g^2 - \alpha^2 r^2}{r^2}\right)^{1/2} = 2\alpha\Delta\theta$$

$$\boxed{\Delta\theta = \frac{(\mu^2 g^2 - \alpha^2 r^2)^{1/2}}{2\alpha r}}$$

Vertical Circular Motion:

Ex



when string becomes vertical

Find

① v_{max} ⇒ T.M.E.C.

Loss in U = Gain in KE

$$mgl = \frac{1}{2}mv_m^2 - 0$$

$$v_{max} = \sqrt{2gl}$$

② $a_c = \frac{v^2}{l} = \frac{2gl}{l} = 2g$

③ $F_t = 0$ (No force in tangential dirⁿ)

④ $a_t = 0$

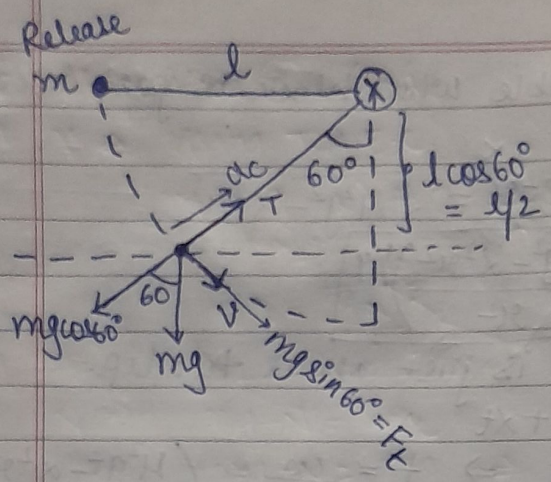
⑤ $a_{net} = 2g$

⑥ Tension in the string: $T - mg = ma_c$

$$T - mg = m(2g)$$

$$\boxed{T = 3mg}$$

Ex



when string is making angle 60° with vertical. Find

① $v \Rightarrow$ Using T.M.E.C.

$$mg\frac{l}{2} = \frac{1}{2}mv^2 - 0$$

$$\boxed{v = \sqrt{gl}}$$

② $a_c = \frac{v^2}{l} = \frac{gl}{l} = g$

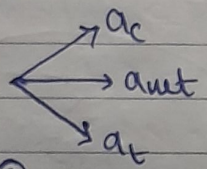
③ $F_t = mg\sin 60° = \frac{\sqrt{3}mg}{2}$

④ $a_t \Rightarrow F_t = ma_t$

$$\frac{\sqrt{3}mg}{2} = ma_t \Rightarrow a_t = \frac{\sqrt{3}g}{2}$$

⑤ $a_{net} = \sqrt{a_c^2 + a_t^2}$

$$= \sqrt{g^2 + \frac{3g^2}{4}} = \sqrt{\frac{7g^2}{4}} = \frac{\sqrt{7}g}{2}$$

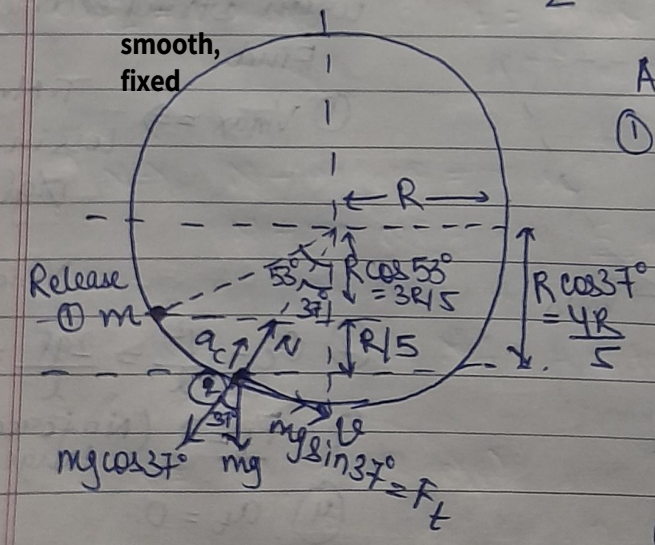


⑥ Tension in the string:

$$T - mg\cos 60° = ma_c$$

$$= T - \frac{mg}{2} = mg \Rightarrow \boxed{T = \frac{3mg}{2}}$$

Ex



At position ②, find:

① $v \Rightarrow$ Using T.M.E.C.

$$mg\frac{R}{5} = \frac{1}{2}mv^2$$

$$\boxed{v = \sqrt{\frac{2gR}{5}}}$$

② $a_c = \frac{v^2}{R} = \frac{2gR}{5R} = \frac{2g}{5}$

③ $F_t = mg\sin 37° = \frac{3mg}{5}$

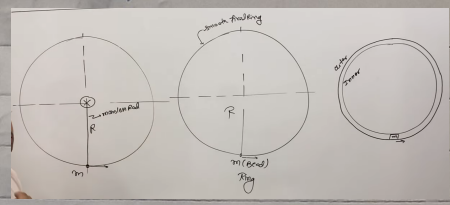
④ $a_t \Rightarrow \frac{3mg}{5} = ma_t$

$$\boxed{a_t = \frac{3g}{5}}$$

⑤ $a_{net} = \sqrt{a_c^2 + a_t^2} = \sqrt{\frac{4g^2}{5} + \frac{9g^2}{5}} = \frac{\sqrt{13}g}{5}$

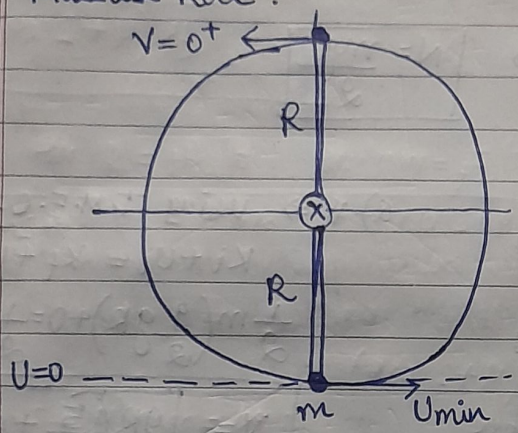
⑥ N⇒ $N - mg\cos 37^\circ = ma_c$

$N - \frac{4mg}{5} = \frac{2mg}{5} \Rightarrow \boxed{N = \frac{6mg}{5}}$



Complete Vertical Circular Motion:

Case I: Massless Rod:



Umin such that particle performs complete vertical C.M.:

using T.M.E.C

$K_i + U_i = K_f + U_f$

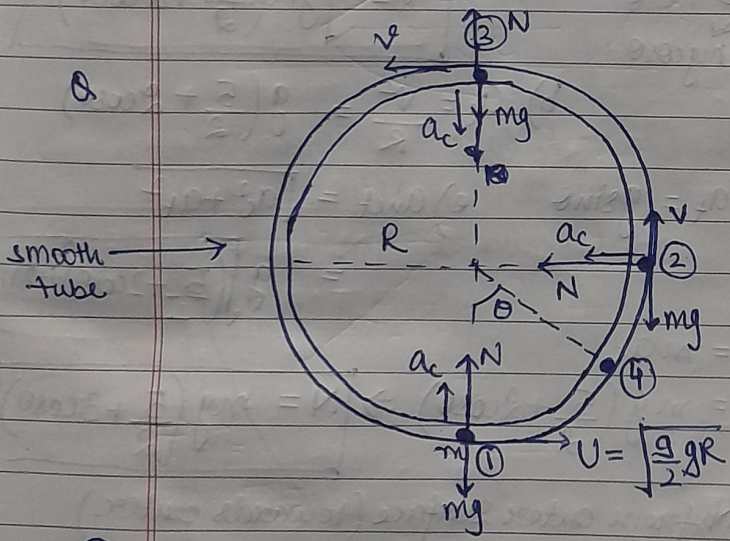
$\frac{1}{2}m\vec{U}_{min} + 0 = 0 + mg(2R)$

$\boxed{U_{min} = \sqrt{4gR}}$

$\therefore \boxed{U \geq \sqrt{4gR}}$

(It is same as massless rod case)

Q



Ⓘ At position ①, find.

a) $a_c = \frac{U^2}{R} = \frac{\frac{9}{2}gR}{R} = 4.5g$

b) $F_t = 0$ c) $a_t = 0$ d) $a_{net} = 4.5g$

e) N⇒ $N - mg = ma_c$

$N - mg = 4.5mg$

$\boxed{N = 5.5mg}$

Ⓘ At position ②, find

a) V⇒ using T.M.E.C

$\frac{1}{2}m\left(\frac{9}{2}gR\right) + 0 = \frac{1}{2}mv^2 + mgR$

$\boxed{V = \sqrt{\frac{5}{2}gR}}$

b) $a_c = \frac{V^2}{R} = \frac{5}{2}g$

c) $F_t = mg$ d) $a_t = g$

e) $a_{net} = \sqrt{\frac{25}{4}g^2 + g^2} = \frac{\sqrt{29}}{2}g$

④ N⇒ $N = ma_c \Rightarrow N = \frac{5}{2}mg$

$\therefore \boxed{N = 2.5mg}$

(III) At position (3), find

a) $V \Rightarrow$ using T.M.E.C.

$$\frac{1}{2}m\left(\frac{g}{2}gR\right)+0=\frac{1}{2}mv^2+mg2R$$

$$\boxed{V=\sqrt{\frac{gR}{2}}}$$

b) $a_c=\frac{v^2}{R}=\frac{g}{2}$

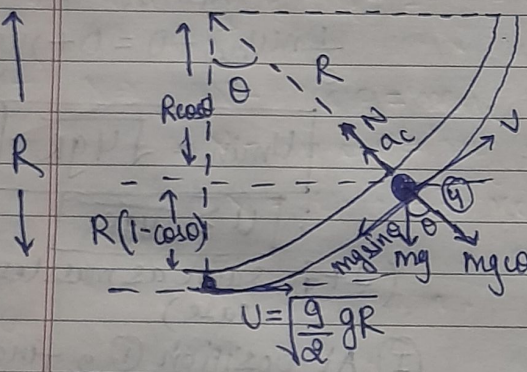
c) $F_t=0$ d) $a_t=0$

e) $a_{net}=\frac{g}{2}$

f) $N \Rightarrow$ $mg-N=ma_c$

$$mg-N=\frac{mg}{2} \Rightarrow \boxed{N=\frac{mg}{2}}$$

(IV) At position (4), find



a) $v \Rightarrow$ using T.M.E.C.

$$K_i+U_i=K_f+U_f$$

$$\frac{1}{2}m\left(\frac{g}{2}gR\right)+0=\frac{1}{2}mv^2+mg(R(1-\cos\theta))$$

$$\boxed{v=\sqrt{gR\left(\frac{5}{2}+2\cos\theta\right)}}$$

b) $a_c=\frac{v^2}{R}=g\left(\frac{5}{2}+2\cos\theta\right)$

c) $F_t=mg\sin\theta$ d) $a_t=g\sin\theta$ e) $a_{net}=\sqrt{a_c^2+a_t^2}$

$$=g\sqrt{\left(\frac{5}{2}+2\cos\theta\right)^2+\sin^2\theta}$$

f) $N \Rightarrow$ $N-mg\cos\theta=ma_c$

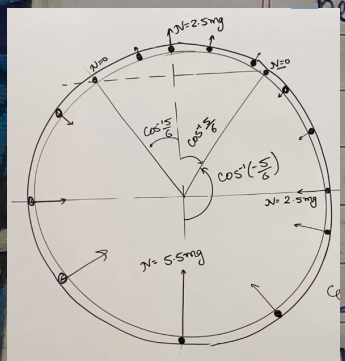
$$N-mg\cos\theta=mg\left(\frac{5}{2}+2\cos\theta\right) \Rightarrow \boxed{N=mg\left(\frac{5}{2}+3\cos\theta\right)}$$

(V) Find angle at which N from outer surface (towards center) becomes zero.

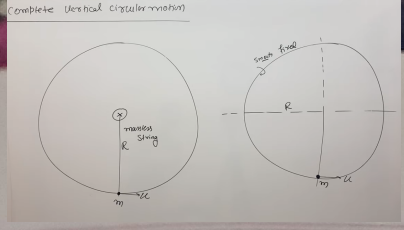
$$N=mg\left(\frac{5}{2}+3\cos\theta\right), \quad N=0 \Rightarrow \frac{5}{2}+3\cos\theta=0$$

$$\Rightarrow \cos\theta=-\frac{5}{6}$$

$$\therefore \boxed{\theta=\cos^{-1}\left(-\frac{5}{6}\right)}$$

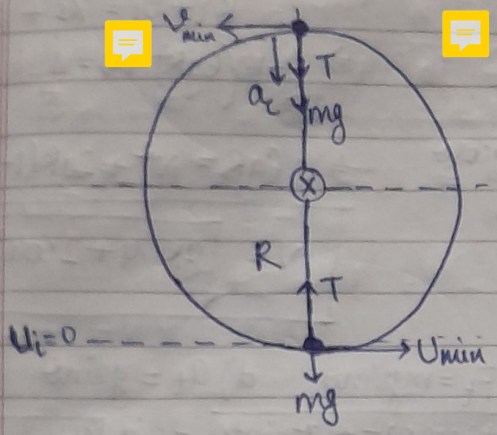


Cases of tube or a ring fixed in a circular wire act same as the case of massless rod because in all these cases the motion is constrained to be in a circular path.



Page No.
Date

Case 2: Massless string (Here particle is not constrained to move in a circle):



V_{min} at highest point:

$$T + mg = \frac{mV_{min}^2}{R}$$

For V_{min}, T should be minimum

$\therefore T = 0$

$$mg = \frac{mV_{min}^2}{R}$$

$$\boxed{V_{min} = \sqrt{gR}}$$

U_{min} such that particle completes vertical circular motion:

Using T.M.E.C $\Rightarrow$ $K_i + U_i = K_f + U_f$

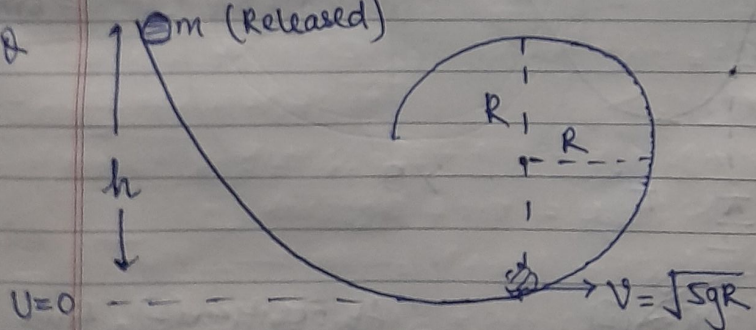
$$= \frac{1}{2}mU_{min}^2 + 0 = \frac{1}{2}m(gR) + mg(2R)$$

$$= U_{min}^2 = 5gR$$

$$= \boxed{U_{min} = \sqrt{5gR}} \quad \therefore \boxed{U \geq \sqrt{5gR}}$$

★ In this case T should not be zero b/w lowest and highest pt, otherwise particle will fall inside. At the highest pt it can be zero for a moment.

Q



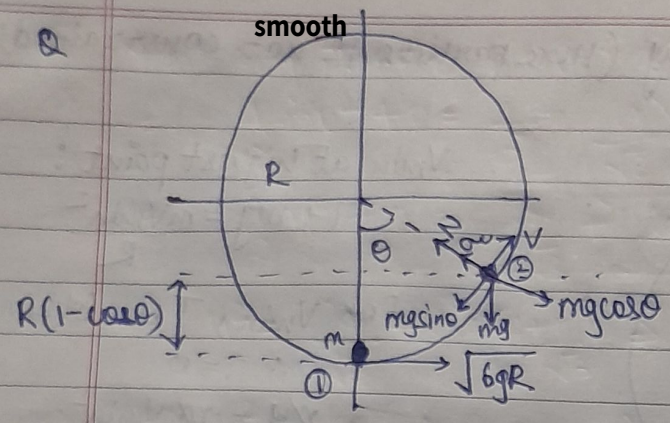
Find minimum value of h so that particle performs complete vertical circular motion on circular track.

Using T.M.E.C $\Rightarrow$ $K_i + U_i = K_f + U_f$

$$0 + mgh = \frac{1}{2}m(5gR) + 0$$

$$\boxed{h = \frac{5R}{2}}$$

Q



At position ②, find

a) $v \Rightarrow$ using T.M.E.C:

$K_i + U_i = K_f + U_f$

$\Rightarrow \frac{1}{2}m(6gR) + 0 = \frac{1}{2}mv^2 + mg(R(1-\cos\theta))$

$\Rightarrow 6gR = v^2 + 2gR(1-\cos\theta)$

$$\therefore \boxed{v = \sqrt{gR(4+2\cos\theta)}}$$

b) $a_c = \frac{v^2}{R} = g(4+2\cos\theta)$ c) $F_t = mg\sin\theta$ d) $a_t = g\sin\theta$

e) $a_{net} = \sqrt{a_c^2 + a_t^2} = \sqrt{g^2(4+2\cos\theta)^2 + g^2\sin^2\theta} = g\sqrt{(4+2\cos\theta)^2 + \sin^2\theta}$

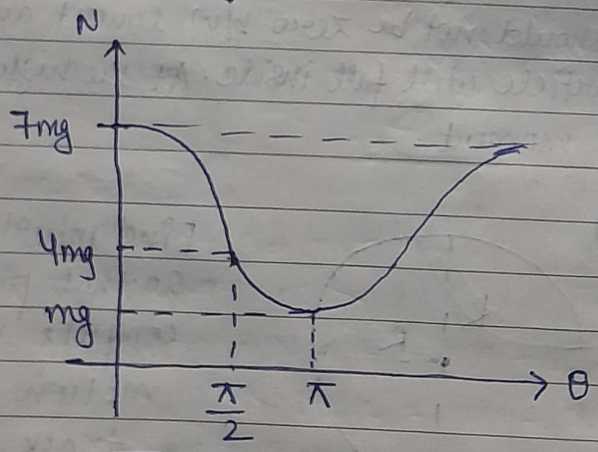
f) $N \Rightarrow$ $N - mg\cos\theta = ma_c$

$N - mg\cos\theta = mg(4+2\cos\theta)$

$N = 4mg + 3mg\cos\theta$

$$\therefore \boxed{N = mg(4+3\cos\theta)}$$

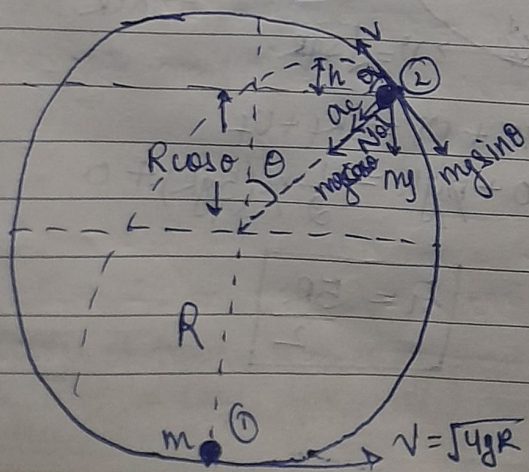
g) Draw N v/s θ graph:



$$\boxed{N_{bottom} - N_{top} = 7mg - mg = 6mg}$$

Q

smooth, fixed circular track



At position ①, find

a) $v \Rightarrow$ using T.M.E.C.

$K_i + U_i = K_f + U_f$

$\frac{1}{2}m(4gR) + 0 = \frac{1}{2}mv^2 + mgR(1+\cos\theta)$

$$\boxed{v = \sqrt{2gR(1-\cos\theta)}}$$

b) $a_c = \frac{v^2}{R} = 2g(1-\cos\theta)$

c) $N \Rightarrow N + mg\cos\theta = ma_c$

$N + mg\cos\theta = m(2g(1-\cos\theta))$

$\boxed{N = 2mg - 3mg\cos\theta}$

d) Find θ, when particle leaves the track.

$N = 0 \Rightarrow 2mg = 3mg\cos\theta$

$\cos\theta = \frac{2}{3} \Rightarrow \boxed{\theta = \cos^{-1}\left(\frac{2}{3}\right)}$

e) Find v of particle when it breaks off contact with track.

$v = \sqrt{2gR(1-\cos\theta)}$

$\boxed{v = \sqrt{2gR\left(1-\frac{2}{3}\right)} = \sqrt{\frac{2gR}{3}}}$

f) What will be the motion of particle when it breaks off the track?

⇒ Projectile motion

g) Find max^m height achieved by the particle from position ①?

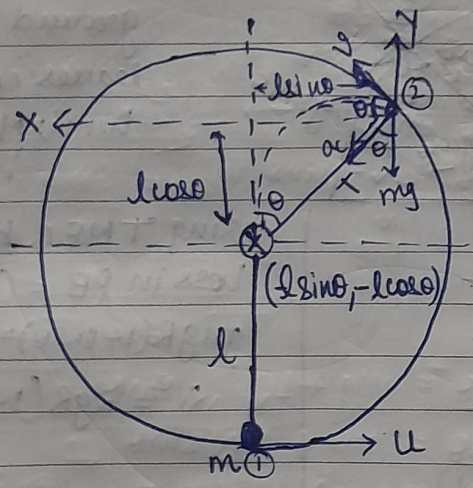
$H = R + R\cos\theta + h$

$= R + R\cos\theta + \frac{v^2\sin^2\theta}{2g} = R + R\left(\frac{2}{3}\right) + \frac{\left(\frac{2gR}{3}\right)\left(\frac{5}{9}\right)}{2g}$

$= R + \frac{2R}{3} + \frac{5R}{27} = \frac{50R}{27}$

$\therefore \boxed{H_{max} = \frac{50R}{27}}$

Q



Find u such that particle strikes the hinge.

At position ②:

$T + mg\cos\theta = \frac{mv^2}{l}$

To loose contact ⇒ $T = 0$

$mg\cos\theta = \frac{mv^2}{l}$

$v = \sqrt{lg\cos\theta}$ — ①

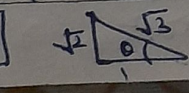
using Eqn of trajectory: $y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$

$\Rightarrow -l\cos\theta = \frac{l\sin^2\theta}{\cos\theta} - \frac{gl^2\sin^2\theta}{2v^2\cos^2\theta}$

$= -\cos\theta = \frac{\sin^2\theta}{\cos\theta} - \frac{gl\sin^2\theta}{2v^2\cos^2\theta} \Rightarrow -\cos\theta = \frac{\sin^2\theta}{\cos\theta} - \frac{gl\sin^2\theta}{2lg\cos^3\theta}$ (using eq(1))

$= -2\cos^4\theta = 2\cos^2\theta\sin^2\theta - \sin^2\theta \Rightarrow \sin^2\theta = 2\cos^2\theta(\cos^2\theta + \sin^2\theta)$

$= \frac{\sin^2\theta}{\cos^2\theta} = 2 \Rightarrow \boxed{\tan\theta = \sqrt{2}}$



$\therefore \cos\theta = \frac{1}{\sqrt{3}} \therefore \boxed{v = \frac{lg}{\sqrt{3}}}$

Now using T.M.E.C b/w ① and ②: $K_i + U_i = K_f + U_f$

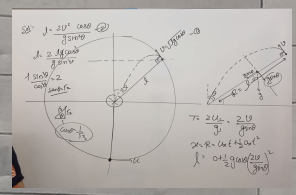
$$= \frac{1}{2}mv^2 + 0 = \frac{1}{2}m\left(\frac{lg}{\sqrt{3}}\right) + mg\left(l(1+\cos\theta)\right)$$

$$= \frac{1}{2}v^2 + 0 = \frac{lg}{2\sqrt{3}} + lg\left(1+\frac{1}{\sqrt{3}}\right) \Rightarrow \frac{v^2}{2} = \frac{lg}{2\sqrt{3}} + lg\left(\frac{\sqrt{3}+1}{\sqrt{3}}\right)$$

$$= v^2 = \frac{lg}{\sqrt{3}} + 2lg\left(1+\frac{1}{\sqrt{3}}\right) \Rightarrow v^2 = \frac{lg}{\sqrt{3}} + lg\left(2+\frac{2}{\sqrt{3}}\right)$$

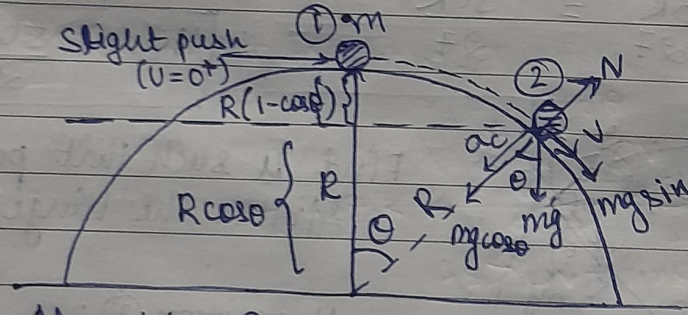
$$= v^2 = lg\left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}} + 2\right) = lg\left(2+\sqrt{3}\right)$$

$$\therefore \boxed{v = \sqrt{lg(2+\sqrt{3})}}$$



Alternate: Final motion can be thought of as projectile motion down the incline. Use range formula to get relation b/w v and θ and then solve.

Q



Find height from ground when particle leaves contact with hemisphere.

At position ②,

$$mg\cos\theta - N = \frac{mv^2}{R}$$

For no contact: N = 0

$$mg\cos\theta = \frac{mv^2}{R}$$

$$v^2 = Rg\cos\theta \quad -①$$

Using ⊤MEC b/w ① and ②

Loss in PE = Gain in KE

$$mgR(1-\cos\theta) = \frac{1}{2}mv^2 - 0$$

$$v^2 = 2gR(1-\cos\theta) \quad -②$$

Using ① and ② ⇒

$$Rg\cos\theta = 2Rg - 2Rg\cos\theta$$

$$3Rg\cos\theta = 2Rg$$

$$\cos\theta = \frac{2}{3} \Rightarrow \theta = \cos^{-1}\left(\frac{2}{3}\right)$$

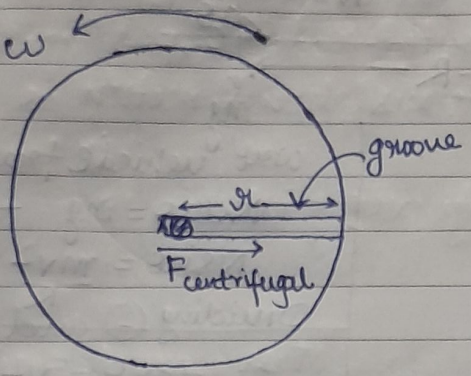
Height from ground = $R\cos\theta = \frac{2R}{3}$

when particle breaks contact

Centrifugal Force:

→ It is a pseudo force (not an actual force, it is a fictitious force; just a mathematical term)

→ It is applied by non-inertial frame of reference.
accelerated observer (performing circular motion)



From frame of ref of disc the ball will move straight from the groove. Pseudo force $\vec{F}_{centrifugal}$ acts radially outwards.

Magnitude of $\vec{F}_{centrifugal} = m\omega^2 r = \frac{mv^2}{r} = m\omega v$

Direction = Radially outwards, or along $\hat{r}$

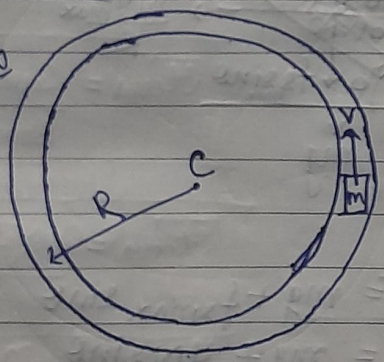
Vector Form: $\boxed{\vec{F}_{centrifugal} = m\omega^2\vec{r}}$

$\boxed{\vec{F}_{centrifugal} = -m(\vec{\omega}\times\vec{v}) = m(\vec{v}\times\vec{\omega})}$

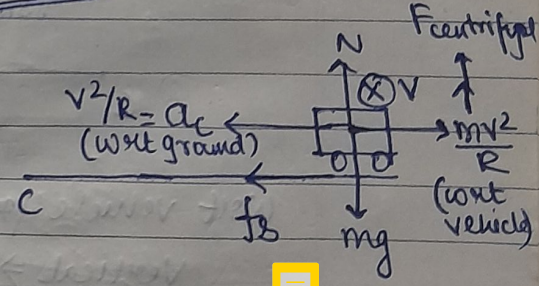
Motion of a vehicle on circular track:

① By Friction:

Top view



Side view



wrt Ground Frame:

$f_s = \frac{mv^2}{R}$

We know, $f_s \le f_{lim}$

$\frac{mv^2}{R} \le \mu mg$

$\boxed{v \le \sqrt{\mu Rg}}$

wrt vehicle:

$N = mg$

$f_s = \frac{mv^2}{R}$

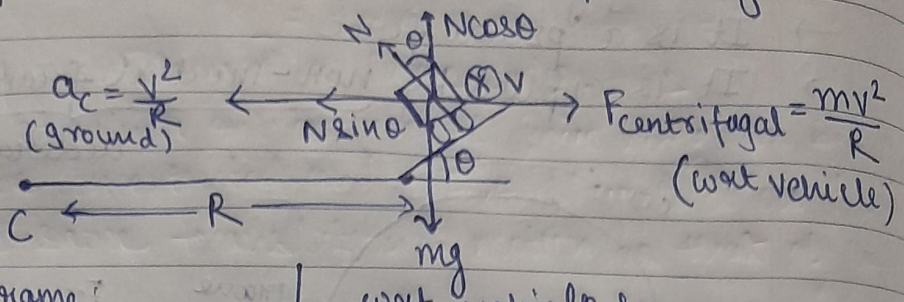
We know, $f_s \le f_{lim}$

$\frac{mv^2}{R} \le \mu mg$

$\boxed{v \le \sqrt{\mu Rg}}$

② By Banking of Road: $\theta \rightarrow$ Angle of banking

with no friction



wrt ground frame:

$N\cos\theta = mg$ —①

$N\sin\theta = \frac{mv^2}{R}$ —②

Dividing ② and ①

$\tan\theta = \frac{v^2}{Rg}$

$\boxed{v = \sqrt{Rg\tan\theta}}$

wrt vehicle frame:

$N\cos\theta = mg$ —①

$N\sin\theta = \frac{mv^2}{R}$ —②

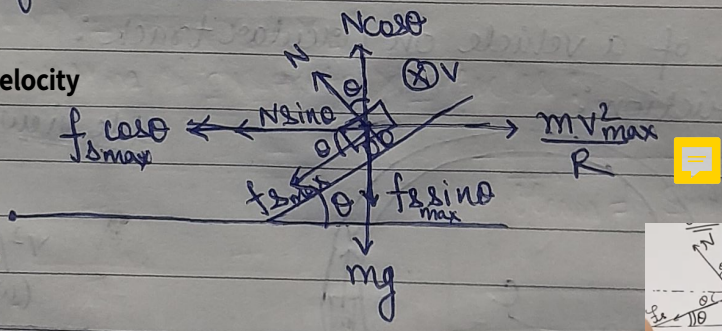
Dividing ② and ①

$\tan\theta = \frac{v^2}{Rg}$

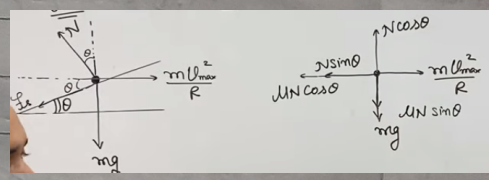
$\boxed{v = \sqrt{Rg\tan\theta}}$

thus, safe negotiation of turn is only possible for a particular speed since the normal is not self-adjusting here like friction was

③ By Banking + Friction:



Case-1: to find maximum velocity



Wrt vehicle frame:

Vertical ⇒ $N\cos\theta = mg + f_{s\max}\sin\theta$

$N\cos\theta = mg + \mu N\sin\theta$

$N(\cos\theta - \mu\sin\theta) = mg$ —①

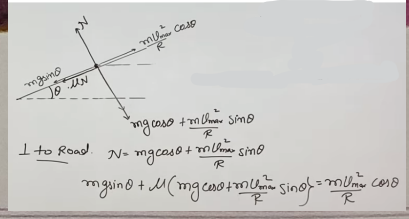
Horizontal ⇒ $N\sin\theta + f_{s\max}\cos\theta = \frac{mv^2_{max}}{R}$

$N(\sin\theta + \mu\cos\theta) = \frac{mv^2_{max}}{R}$ —②

Divide ② by ①

$$\boxed{v_{max} = \sqrt{\frac{Rg(\sin\theta + \mu\cos\theta)}{(\cos\theta - \mu\sin\theta)}} = \sqrt{\frac{Rg(\tan\theta + \mu)}{(1 - \mu\tan\theta)}}}$$

★ If $v > v_{max} \Rightarrow$ car slips outwards



Page No.
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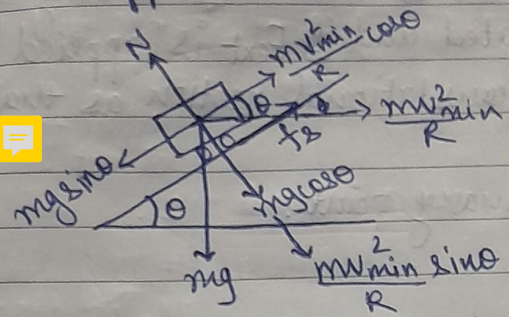
Now we need to find V_{min} : If V is min^m then $\frac{mv^2}{R}$ will be min^m, so component of mg along road ($mg\sin\theta$) will try to slip the car inwards so friction will act towards the opposite direction to prevent this.



$$mg\sin\theta = \mu N + \frac{mV_{min}^2}{R}\cos\theta \quad \text{①}$$

$$N = mg\cos\theta + \frac{mV_{min}^2}{R}\sin\theta \quad \text{②}$$

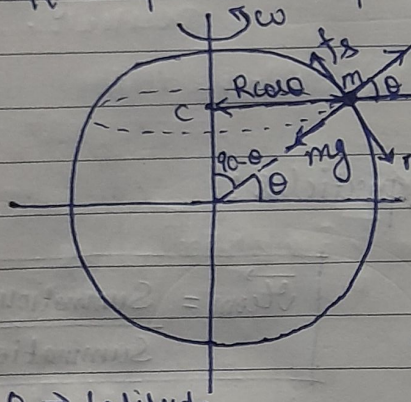
Put ② in ①

$$mg\sin\theta = \mu\left(mg\cos\theta + \frac{mV_m^2}{R}\sin\theta\right) + \frac{mV_m^2}{R}\cos\theta$$

★ If $V < V_{min} \Rightarrow$ car slips inwards

$$\boxed{V_{min} = \sqrt{\frac{Rg(\sin\theta - \mu\cos\theta)}{\cos\theta + \mu\sin\theta}} = \sqrt{\frac{Rg(\tan\theta - \mu)}{1 + \mu\tan\theta}}}$$

Effect of rotation of earth on 'g' :



$$mg = N + m\omega^2 R\cos^2\theta$$

$$\boxed{N = mg - m\omega^2 R\cos^2\theta}$$

↳ Weight apparent

Min weight ← $(W_{equator})_{app} \Rightarrow \theta = 0° \Rightarrow mg - m\omega^2 R$

Max weight ← $(W_{pole})_{app} \Rightarrow \theta = 90° \Rightarrow mg$

Now, $N = mg - m\omega^2 R\cos^2\theta = m\,g_{eff}$

$$\boxed{g_{eff} = g - \omega^2 R\cos^2\theta}$$

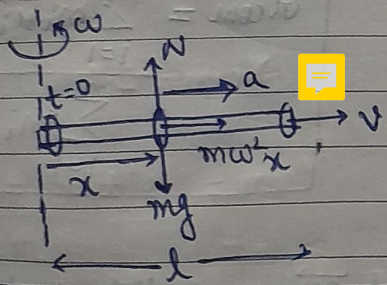
$$\boxed{\begin{aligned} g_{equator} &= g - m\omega^2 R \\ g_{pole} &= g \end{aligned}}$$

θ ⇒ latitude

90-θ ⇒ Co-latitude

$\omega = \frac{2\pi}{24\times60\times60}$ rad/s

Q



Find velocity of bead when it reaches the end of the rod.

Wrt rod (Rotating Frame of ref):

$$m\omega^2 x = ma$$

$$\omega^2 x = a \neq \text{const}$$

$$v\frac{dv}{dx} = \omega^2 x$$

$$\int_0^v v\,dv = \omega^2 \int_0^l x\,dx \Rightarrow \boxed{v = \omega l}$$

Note: From ground frame ↳ Motion will look like a spiral path